About Me

Accredited Bigdata Grade A Trainer

Conducted 1200+ training sessions on Bigdata & Hadoop, Spark, Cassandra, Kafka for 1000s of participants from more than 100 fortune 500 companies with an average of 95% success results

Veer Anjaneya Raju Nagaraju

CERTIFICATIONS

- Accredited Bigdata Trainer
- Accredited PMP & MSP Practitioner
- ITIL V3 Certified
- MS(Computer Science) from Columbus University

TRAINING

- Conducted corporate sessions, online and classroom sessions in across India, APAC and North America with 10 years of teaching experience in designing programs, content development, and conducting technology workshops

EXPERIENCE

- 27 years of total IT experience with 9 years in USA

ROLE

- Coach and Consultant on Bigdata & Hadoop, Spark, Kafka, Cassandra, mongoDB, NiFi, ELK and Talend..
- Online trainer for various famous training providers
- Bigdata Consultant for various corporates

Hosted many webinars on Bigdata Technologies both on Development and Administration as Bigdata Expert.
Was invited as Guest Speaker at various Engineering colleges and technology conferences.
Published articles on emerging technologies on news papers and on TV channels

TEKCRUX
ReDefine IT

BIG DATA & HADOOP OVERVIEW

BIG DATA & HADOOP OVERVIEW

Agenda

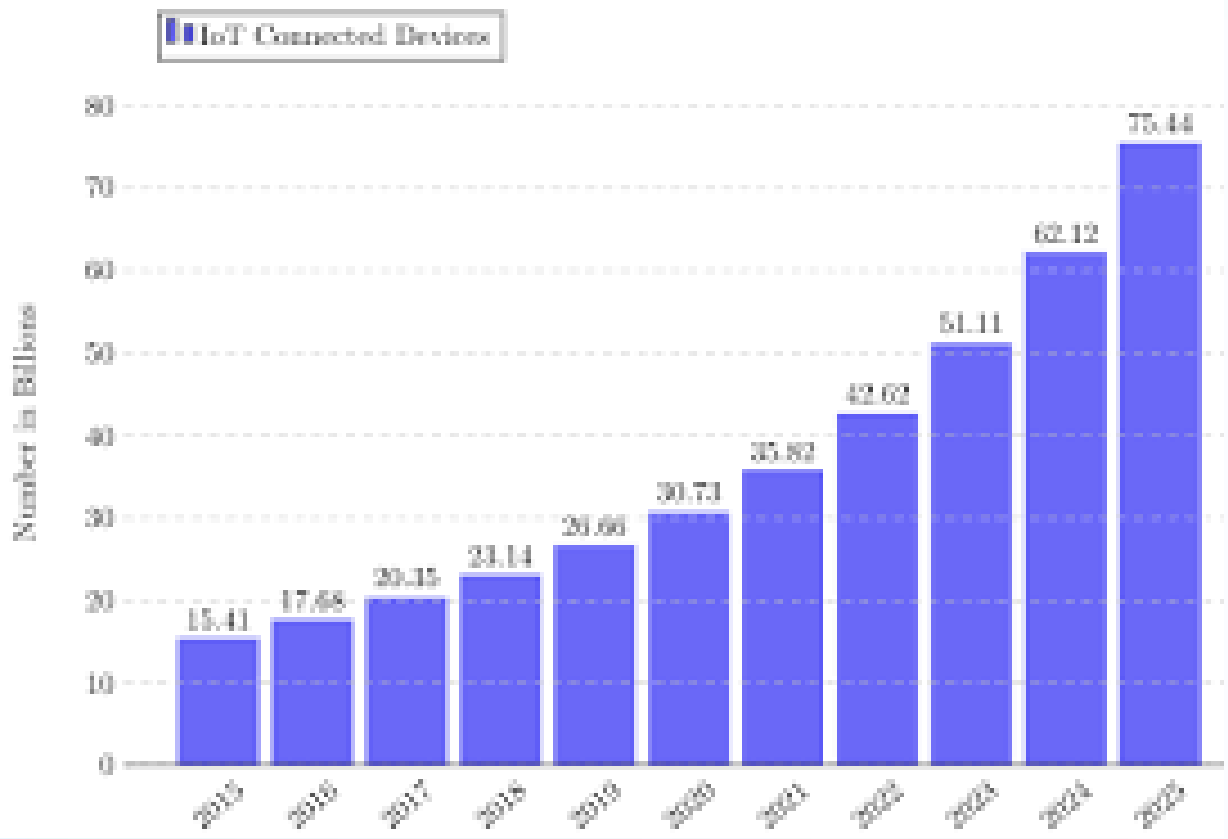
- ✓ Data Explosion
- ✓ Understanding Big Data
- ✓ Big Data Challenges
- ✓ Big Data Opportunity
- ✓ Hadoop as a solution to Big Data
- ✓ Advantages of Hadoop
- ✓ Hadoop Ecosystem

Drivers of Data Explosion – Smart Devices

As devices such as phones and house appliances getting smarter, they also generate huge amount of data.

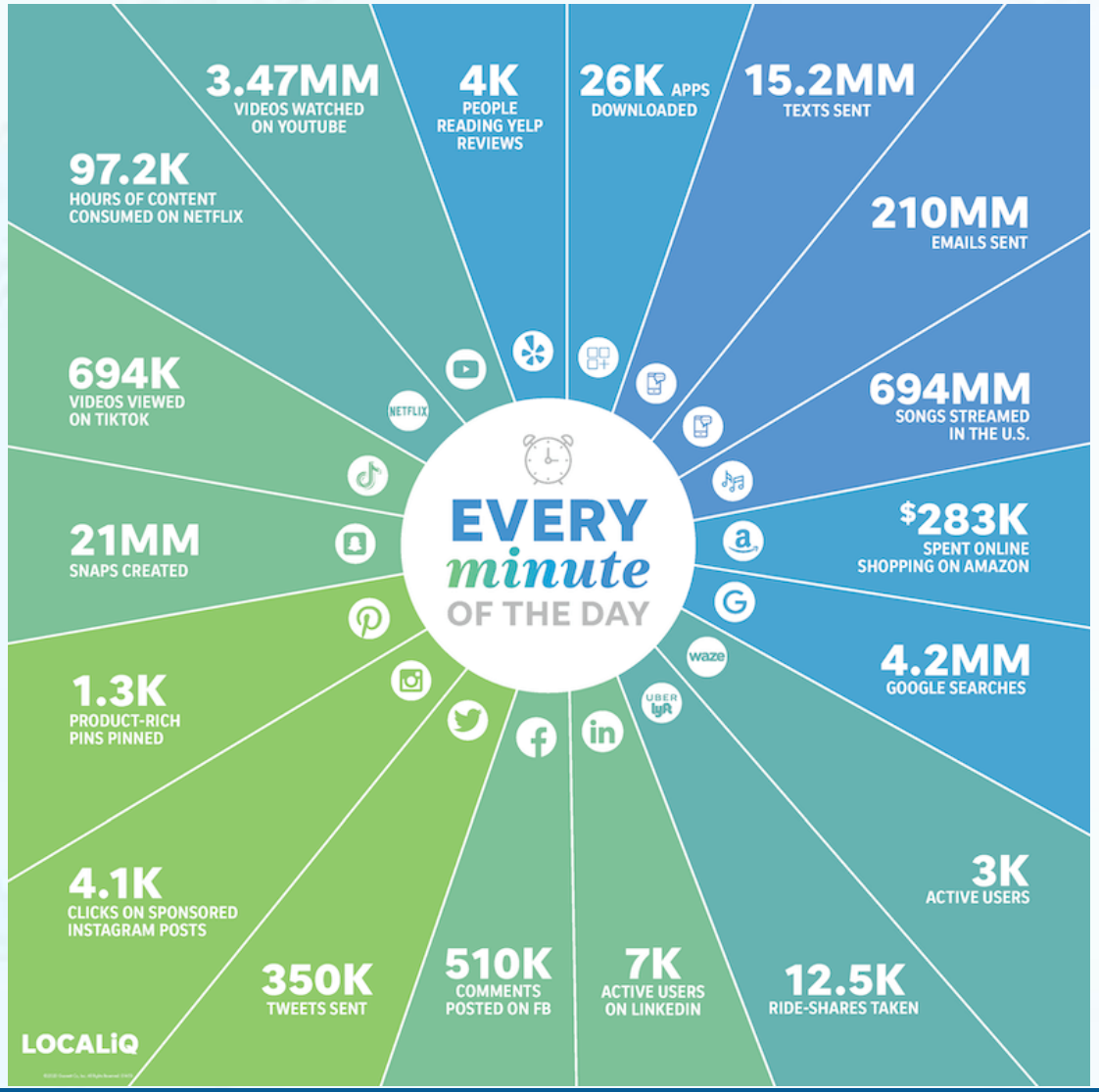
- Smart Phones
- Smart Watches
- Smart Appliances
- Self Driving Cars

Drivers of Data Explosion - IoT



By 2025, there will be 75 billion IoT devices i.e. roughly *8 things online per person* causing huge amount of data getting generated.

Drivers of Data Explosion – Social Media



This Is What Happens In An

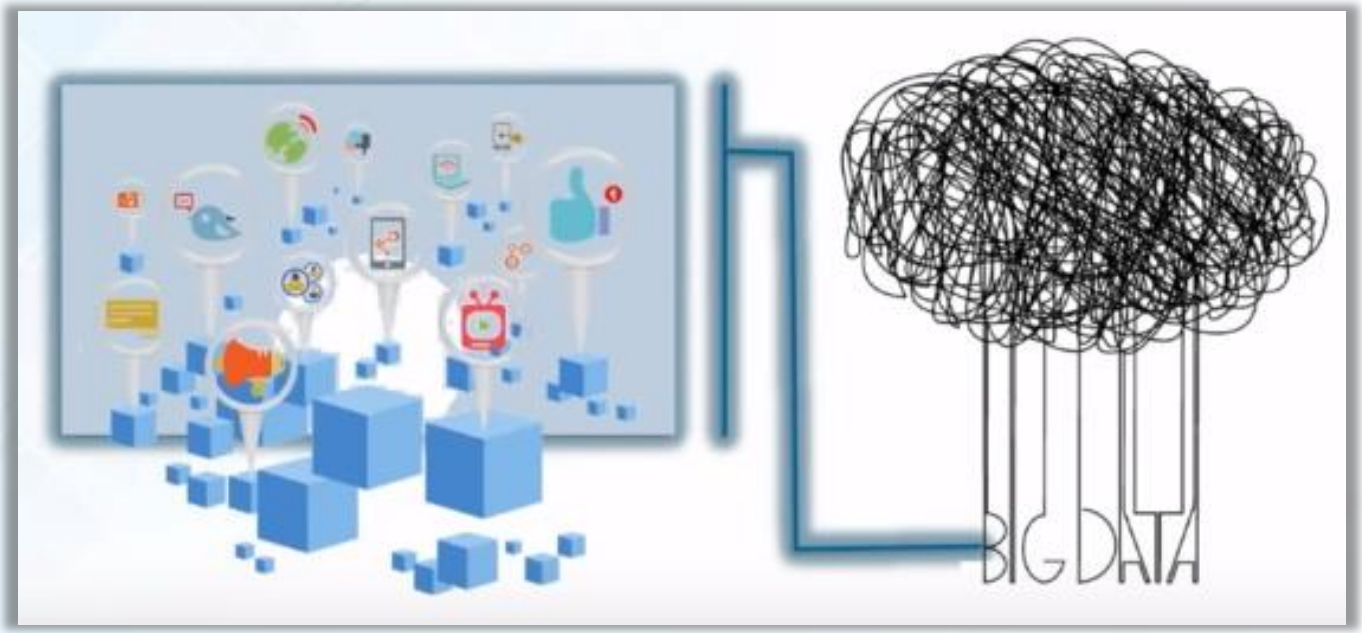
Internet Minute

2022

Drivers of Data Explosion - Others

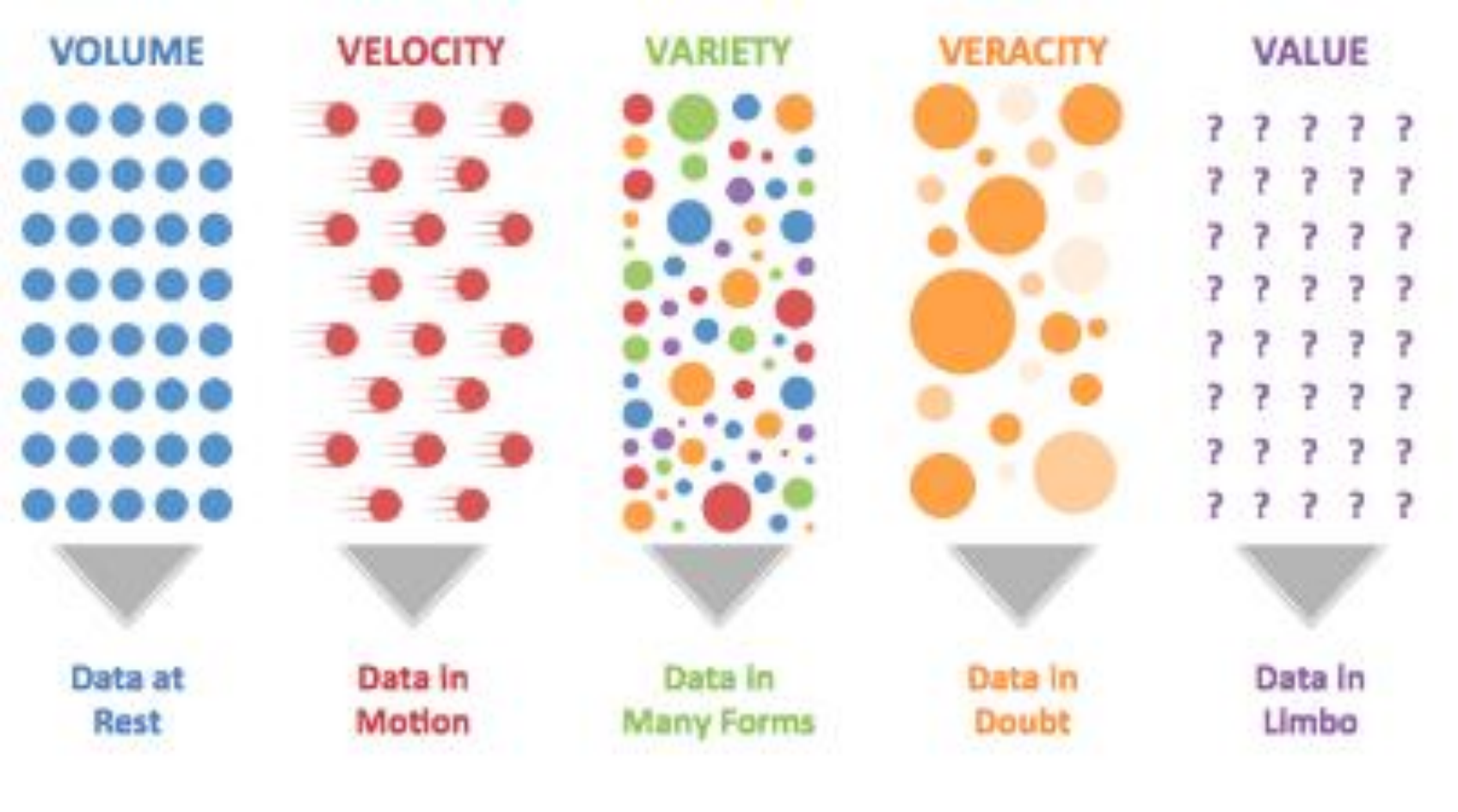
What is Big Data ?

Big Data is a term for a collection of data sets so **large and complex**, that it becomes **difficult to store and process** using on-hand database management tools or traditional data processing applications



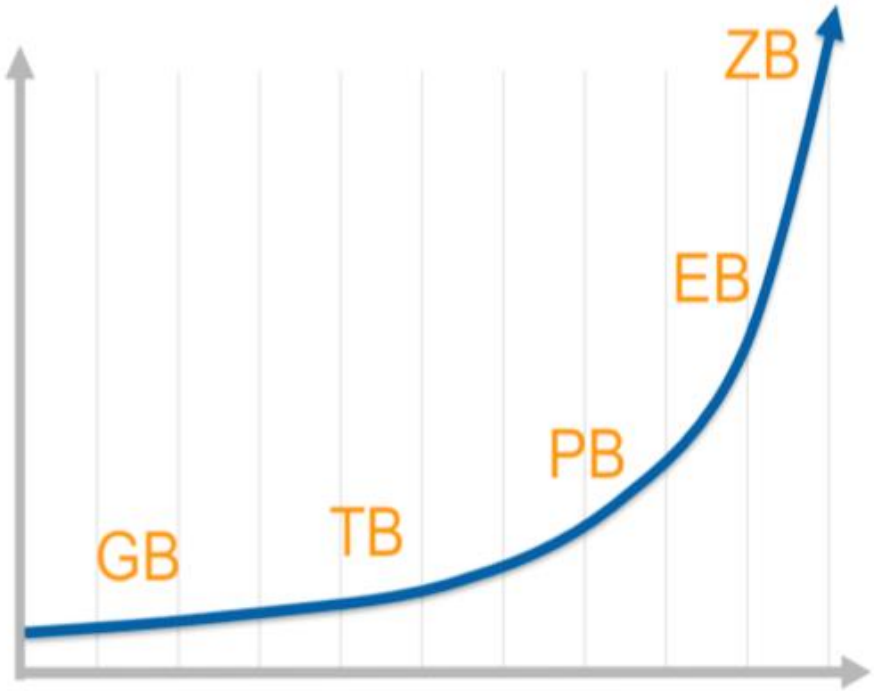
How do we classify some data as Big Data?

5 Vs of Big Data



Volume

1 Volume



Research from IDC shows that unstructured content accounts for 95% of all digital information, with estimates of compound annual growth at 65%.

By 2020, IDC predicts the volume of digital data will have reached 40,000 Exabytes (EB) or 40 Zettabytes (ZB).

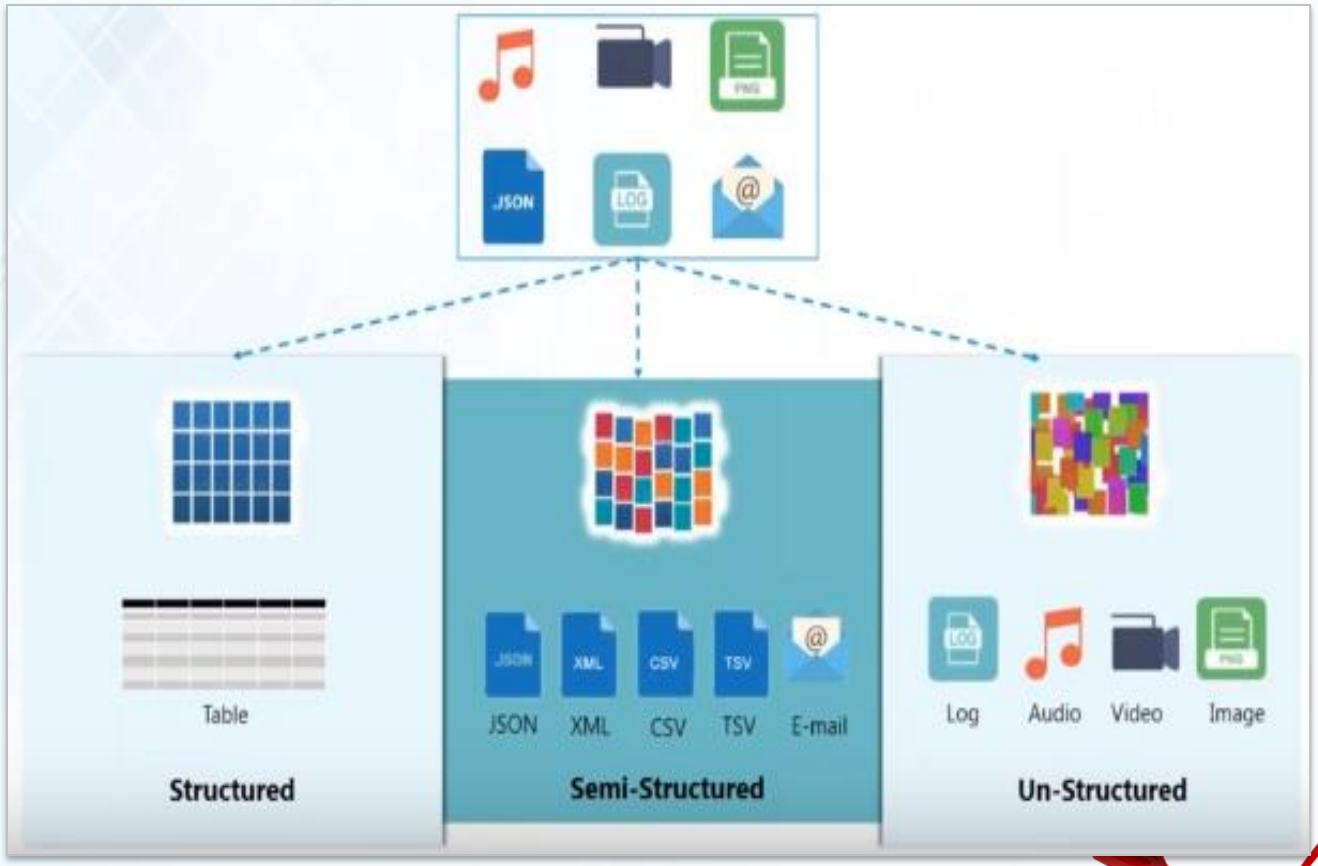
Volume Growth of Unstructured Data

Variety

1 Volume

2 Variety

Data comes in different formats:
Structured, Semi-structured and Unstructured



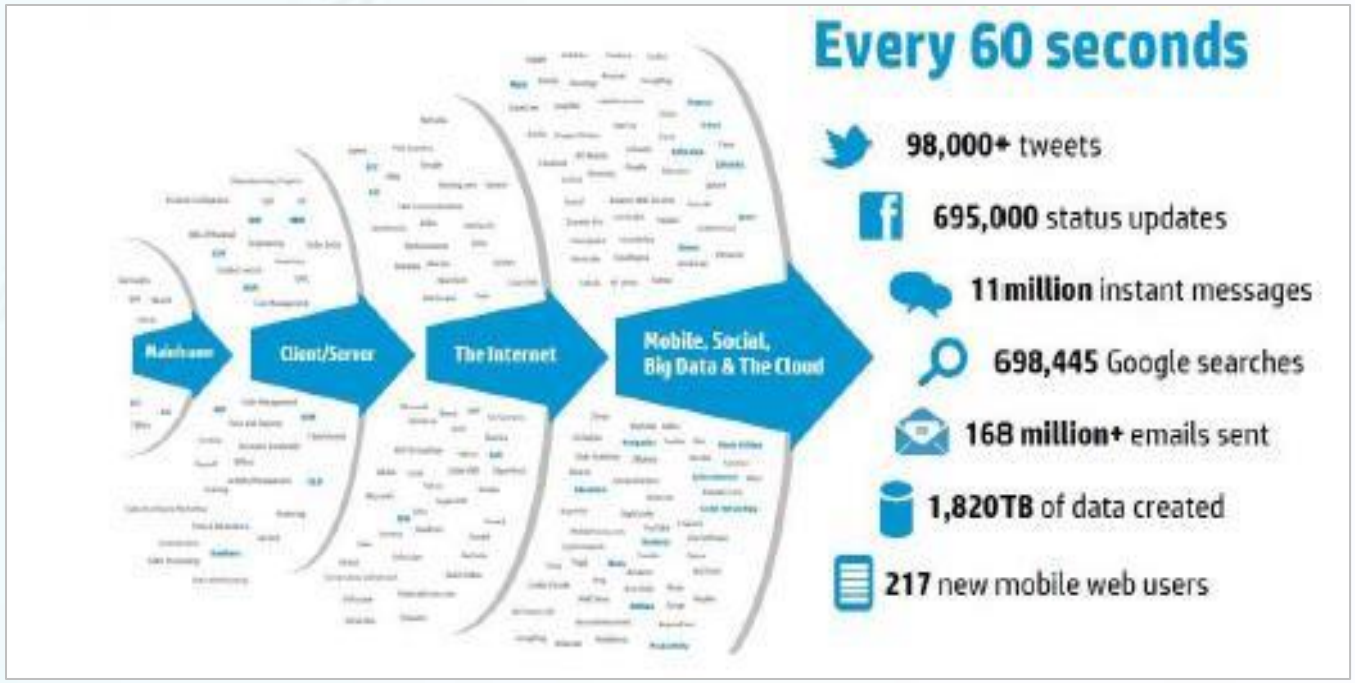
Velocity

1 Volume

2 Variety

3 Velocity

Data is being generated at a very high rate



Value

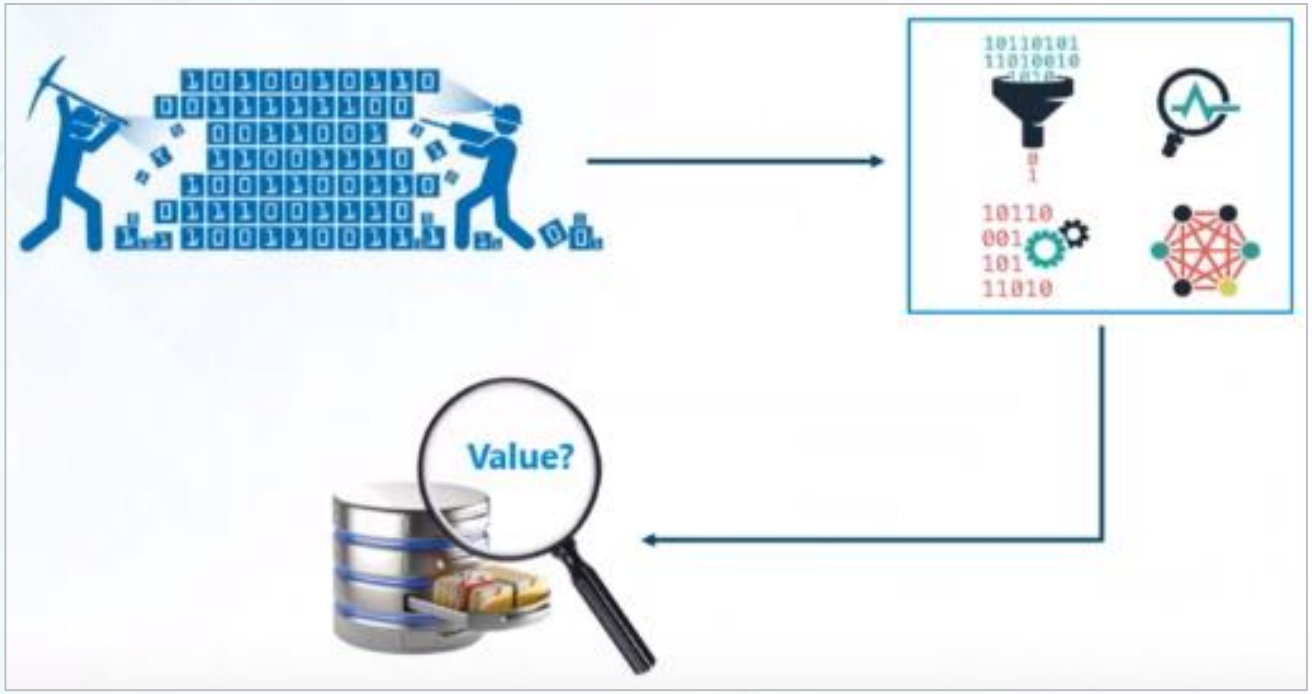
1 Volume

2 Variety

3 Velocity

4 Value

Value refers to the extraction of correct meaning out of the data.



Veracity

1. Volume
2. Variety
3. Velocity
4. Value
5. Veracity

Veracity refers to correctness of data. Data can be ambiguous, uncertain and missing.

Max	Min	Mean	Std. Dev
5.7	4.5	4.8	0.44
10.5	?	11.2	0.32
75000	5.4	9.5	1000
1.5	0.5	?	0.6

Opportunity: Big Data Analytics

Problems with Big Data

1. Storage

- Storing exponentially growing huge datasets using traditional ways is a big problem.

2. Processing complex data

- Processing the data that has complex structure is a problem as the data comes as unstructured and semi-structured forms.

3. Processing data faster

- The data is growing at much faster rate than disk IO speeds. Bringing huge amount of data to the computation unit becomes a bottleneck.

What is the solution to all these Big Data Problems ?

Hadoop is the solution

So lets dive in and look at what Hadoop is all about.

Hadoop

Hadoop is an open-source software framework for storage and processing of datasets on clusters of commodity hardware.

HDFS (Storage)

Allows to store any kind of data in a cluster in a distributed fashion

MapReduce/Spark (Processing)

Allows parallel processing of data stored in HDFS

History of Hadoop

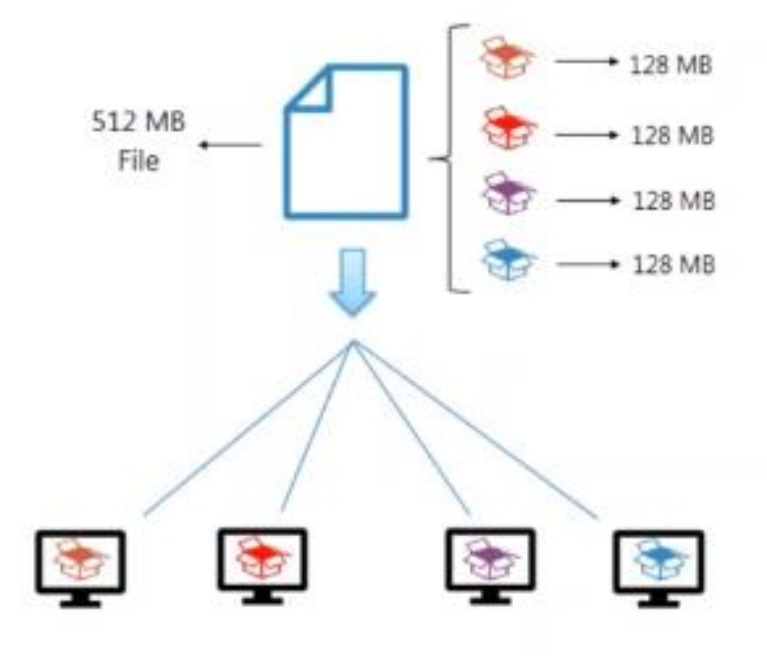
2008	• Hadoop became Apache Top Level Project
2006	• Yahoo! hires Doug Cutting to work on Hadoop with a dedicated team
2005	• Doug Cutting and Nutch team implemented Google's frameworks in Nutch
2004	• Google publishes Google File System (GFS) and MapReduce framework papers

The Hadoop Solution

Problem 1: Storing exponentially growing huge datasets

Solution: HDFS

- Hadoop's storage unit
- Divides files into blocks
- Stores blocks across the cluster
- Replicates blocks as a fail-safe mechanism
- Horizontally Scalable



The Hadoop Solution

Problem 2: Storing unstructured data

Solution: HDFS

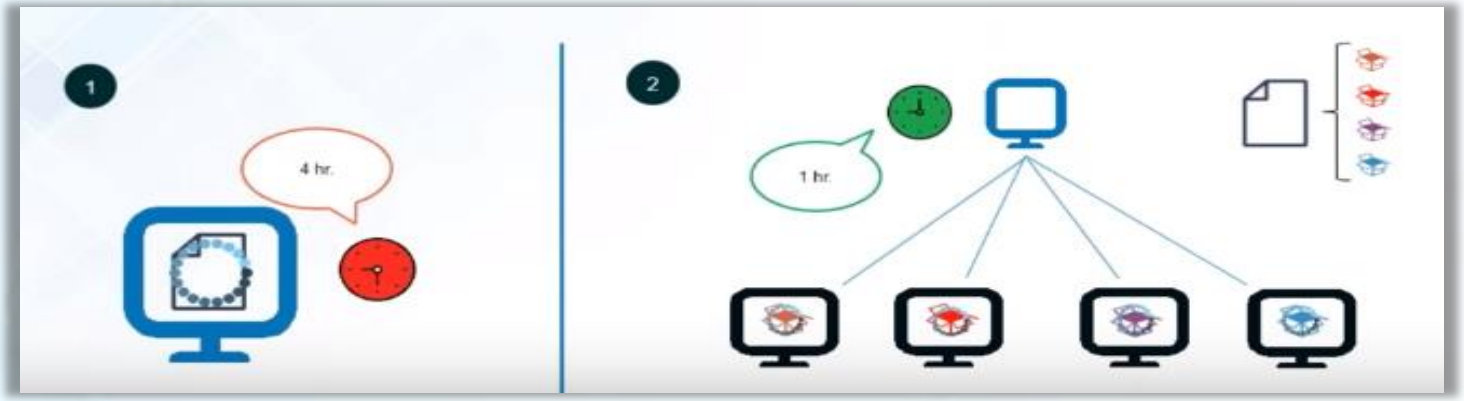
- HDFS allows to store any kind of data – be it structured or not.
- No schema validation done while writing data
- Follows write once and read many times paradigm (WORM)

The Hadoop Solution

Problem 3: Processing data faster

Solution: MapReduce

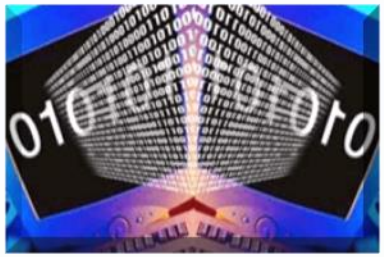
- Provides parallel processing of data present in HDFS
- Allows data to be processed locally on each node making use of its resources.
- Brings processing to the data



Advantages of Hadoop

- Scalable
- Available
- Reliable
- Cost Effective
- Flexible
- Fast
- Fail Safe

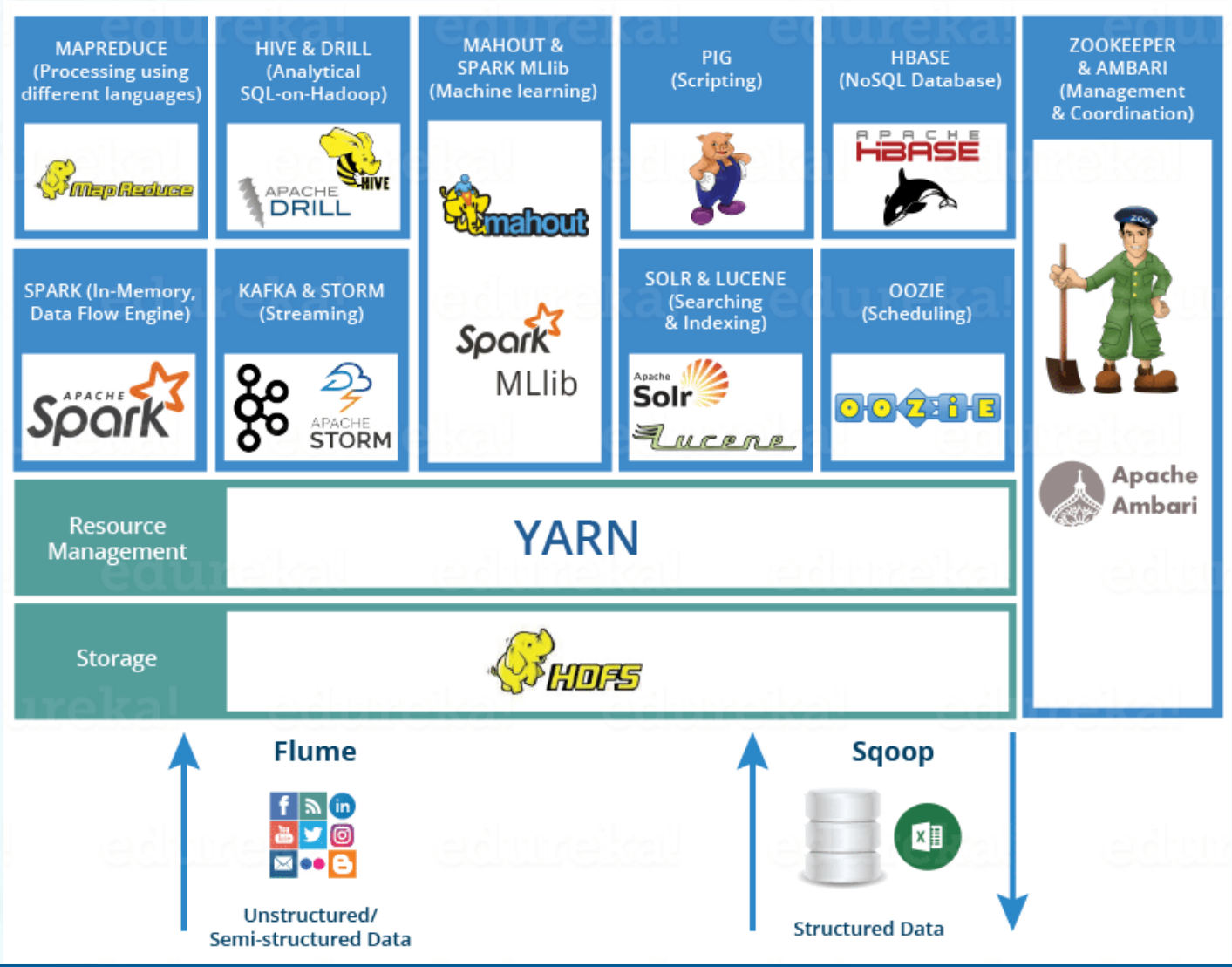
Applications of BigData



Who is using?

AOL
ebay
IBM
twitter
amazon.com
Adobe
facebook
hulu
last·fm
the social music revolution
LinkedIn
YAHOO!

Hadoop Ecosystem



Hadoop Ecosystem

Hadoop distributed storage (HDFS) and distributed processing framework (mapReduce)) for handling big data problems

Apache Hive is a data warehousing tool that provides an SQL like interface to perform data analytics

Apache Spark is an in-memory data processing engine for executing batch, streaming, machine learning and SQL workloads.

Apache HBase is a NoSQL database that uses HDFS for its underlying storage, and supports both batch-style computations using MapReduce and point queries.

Hadoop Ecosystem

Apache Sqoop is a tool for efficiently moving structured data between relational databases and HDFS.

Apache Flume is a distributed service for efficiently collecting, aggregating, and moving large amounts of streaming data.

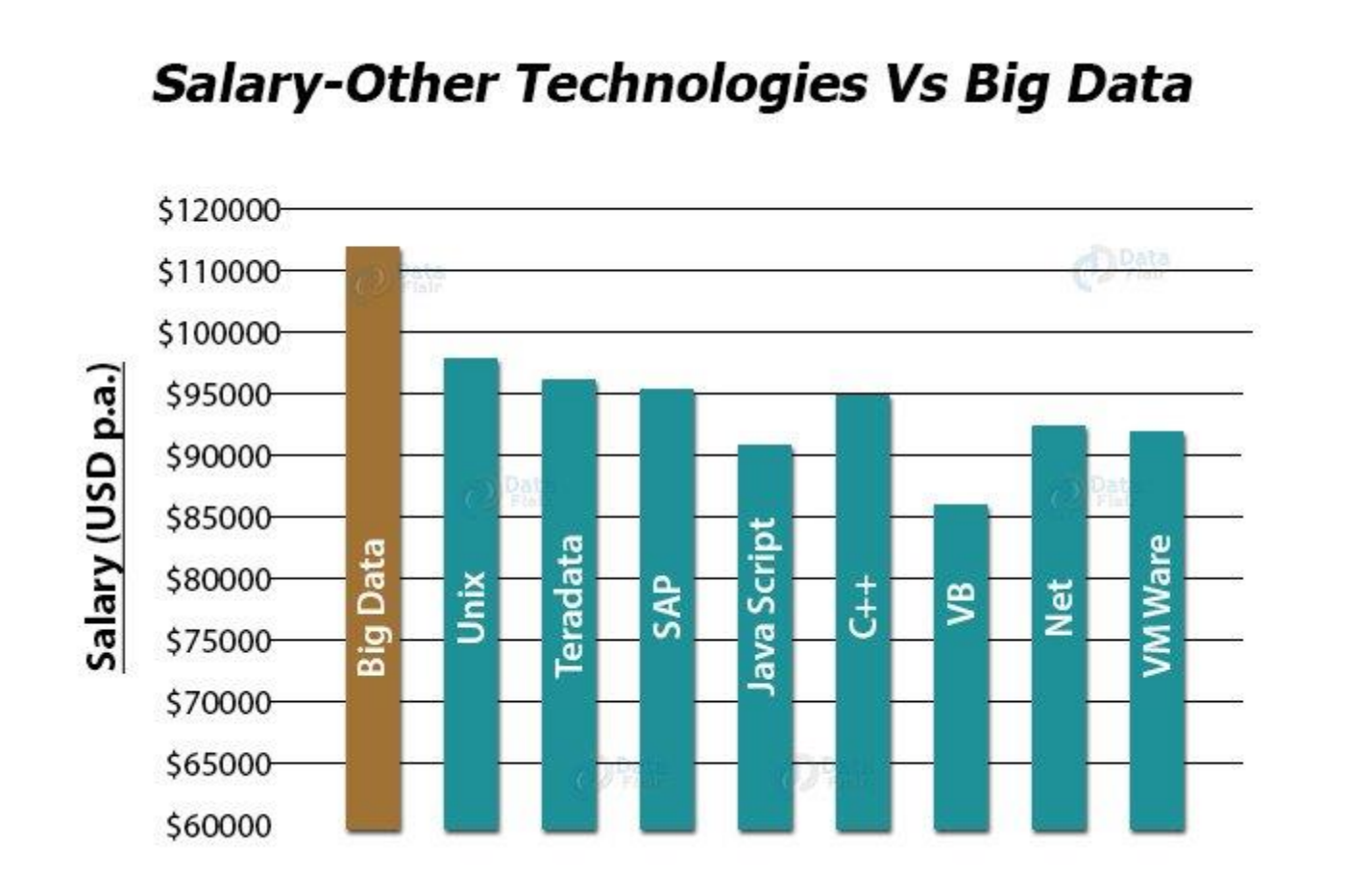
Apache Oozie is a workflow scheduler system to manage Hadoop jobs.

Apache Zookeeper is a distributed coordination service that provides primitives such as distributed locks that can be used for building distributed applications.

Some Big Data Use Cases for Modern Business

- Log Analytics.
- E-Commerce Personalization.
- Recommendation Engines.
- Automated Candidate Placement in Recruiting.
- Insurance Fraud Detection.
- Relevancy and Retention Boost for Online Publishing.
- 360° View of the Customer
- Security Intelligence
- Price Optimization
- Social Media Analysis and Response
- Preventive Maintenance and Support

The future


Salary-Other Technologies Vs Big Data
Salary (USD p.a.)
$120000
$110000
$100000
$95000
$90000
$85000
$80000
$75000
$70000
$65000
$60000
Big Data
Unix
Teradata
SAP
Java Script
C++
VB
Net
VM Ware

TIME FOR QUESTIONS
12
1
2
3
4